

FELICIA CHOO EN YI

Bachelor of Computer Science with Honours (Computer Science) | Phone: +60 11-1328 1812
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PROFESSIONAL SUMMARY

High-achieving Computer Science undergraduate (3.9 CGPA) with a robust foundation in core CS principles, full-stack software development, machine learning, and quality assurance. Proven experience engineering cross-platform desktop/mobile applications, integrating intelligent AI features, and developing robust automation scripts and backend systems. A versatile problem solver skilled in Python, Java, JavaScript, React, and SQL, seeking to leverage strong analytical capabilities and software engineering best practices across diverse technical roles.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, SQL, C, Scratch
Software Development: Object-Oriented Programming (OOP), Data Structures & Algorithm Design, Software Integration, DLL Invocation, Process Automation, API Integration
Web & Application Development: React.js, Node.js, Responsive Web Design (Mobile & Desktop), Client-Server Architecture, MVC Architecture, Java Socket Programming, REST API Development
AI & Machine Learning: Predictive Modeling, Time-Series Forecasting, XGBoost, Random Forest, Decision Trees, LSTM/GRU, Feature Importance & Model Interpretation, AI Feature Engineering & Design, Model Evaluation & Validation
UI/UX Design: Wireframing, User Research, User Personas, HCI Principles, Usability Testing, System Usability Scale (SUS)
Databases: MongoDB, Oracle Database, SQL, Relational Database Design, Database Modeling, CRUD Operations
Cybersecurity & Network Security: IDS/IPS, Snort, Suricata, Wazuh, Splunk, Network Traffic Analysis, Security Log Analysis, Custom IDS/IPS Rule Development, Security Event Monitoring, HIDS, NIDS, Incident Response & Security Investigation
Networking: Network Analysis (PingPlotter), TCP/IP Networking, Network Programming, Routing & IP Configuration, Virtual Network Segmentation, Packet & Traffic Analysis, VMware Virtual Network Configuration
Testing & Hardware: Optical Transceiver Testing (ParTest), Digital Communications Analyzer (DCA), Optical Spectrum Analyzer (OSA), Optical Power Meter, Embedded Systems (TinkerCAD), Hardware & Network Troubleshooting
Tools & Platforms: GitHub, Visual Studio Code, Eclipse, VMware, Linux/Ubuntu, Kali Linux, Windows
Soft Skills: Communication, Technical Teaching & Lesson Design, Problem Solving, Team Collaboration, Project Planning, Critical Thinking, Self-Learning, Technical Documentation

WORK EXPERIENCE

INVITED SPEAKER, Google for Education | APAC Higher Education Leader Series 2026, Jun 2026
Driving Malaysia's Higher Education Transformation: From AI Policy to Practice

- Co-led an executive live demonstration alongside Prof. Dr. Ts. Nurfadhlina binti Mohd Sharef for an international audience of university presidents, provosts and CIOs, demonstrating practical applications of AI in higher education.
- Demonstrated Gemini and NotebookLM for workflow structuring, data visualisation and AI-assisted code generation in retail analytics.
- Showcased prompt engineering with Gemini Canvas to generate dashboard layouts automatically.
- Connected Malaysia's national AI strategic vision with practical classroom applications and student learning outcomes.

Intern, Testing and Development Team, Coherent Sdn.Bhd | *Jul 2026 – Oct 2026*

Standalone Traceview & DCA Images Cleanup Service

- Automated file-system cleanup by developing a standalone executable that identifies and deletes Traceview and DCA image files older than 30 days.
- Implemented file-type filtering to remove unnecessary files while retaining required formats such as .txt.
- Improved file-management efficiency by automating outdated-file removal and reducing the need for manual cleanup.

Enhanced Auto-Download & Database Update Service

- Extended an existing auto-download service by implementing DLL invocation to support additional functionality.
- Automated database updates after file downloads by integrating SQL queries.
- Improved data synchronisation between downloaded files and the database, reducing manual intervention.

Technical Exposure: Optical Transceiver Testing

- Gained hands-on training in ParTest for optical transceiver testing, test monitoring and basic failure analysis.
- Used a Digital Communications Analyzer (DCA) to examine eye diagrams and signal quality.

- Gained exposure to an Optical Spectrum Analyzer (OSA) and Optical Power Meter for optical signal and power measurements.
- Learned about temperature-dependent module testing using a KLC Temperature Control Unit, temperature chamber, chiller and booster.

Coding Lab Demonstrator, Universiti Putra Malaysia (UPM) | *Part-time*
- Guided students in CCS3014 Advanced Programming lab sessions, supporting coding, debugging and problem-solving.
- Assisted students in CSC3007 Database Application in Agriculture with database design, SQL queries and system implementation.
- Demonstrated object-oriented programming concepts in CCS3101, including inheritance, polymorphism, exception handling and generics.
- Collaborated with lecturers to support practical lab delivery and smooth lab operations.

Coding Teacher (Python, Scratch, Roblox, Minecraft, EduBlock),CodeWhiz Sdn.Bhd | *Part-time*
- Taught programming to children and beginners using Python, Scratch, Roblox, Minecraft and EduBlock.
- Applied game-based projects and hands-on activities to make programming concepts accessible and engaging.
- Developed lesson plans and learning materials tailored to different ages and skill levels.
- Guided students through debugging and problem-solving while tracking their learning progress.

PROJECTS

***Gadget2Go (G2G): Used-Gadget Marketplace Platform* | *Python, React, MongoDB* | Gold Medal, PEDi**
- Developed a responsive frontend prototype for a second-hand gadget marketplace, incorporating AI-based price evaluation, buyer–seller matching and trust-related features to support more informed purchasing decisions.
- Connected the frontend to MongoDB through Python to store platform data.
- Conducted Malaysian market research, competitor analysis, customer segmentation and SWOT analysis to identify opportunities and challenges in the second-hand gadget market.

Sproutly: Smart Gardening Mobile App* | *UI/UX Design, AI Feature Design, HCI
- Designed a user-centred gardening app prototype featuring AI-assisted plant disease assessment, Yes/No follow-up questions, top-three diagnoses with confidence scores, treatment plans and adaptive reminders.
- Conducted user interviews and secondary research, developed three user personas, and created user journey maps, use case diagrams and low- and high-fidelity wireframes using HCI principles.
- Evaluated the prototype with 22 participants using the System Usability Scale (SUS), achieving an average score of 90.91 (Grade A).

NutriTrack: AI-Powered Elderly Nutrition & Health Monitoring App* | *UI/UX Design, AI Feature Design, HCI
- Designed an elderly-focused health monitoring prototype featuring calorie, water and activity tracking, AI food recognition and voice-based meal logging.
- Designed proposed features for personalised dietary recommendations, abnormal nutrient detection, adaptive reminders and alerts connecting elderly users with caregivers or healthcare professionals.
- Applied HCI principles to design an accessible interface with large visual elements and simplified interactions for meal logging, weekly progress tracking and health monitoring.

Sustainable Retail Analytics for Smarter Inventory & Reduced Waste (SDG 12)* | *Python, Machine Learning
- Combined three Walmart datasets containing approximately 421,000 rows and engineered temporal and promotional features to support retail sales prediction.
- Developed and compared regression and classification models, with Decision Tree/Random Forest achieving a reported mean R^2 of 0.942 for weekly sales prediction and Decision Tree achieving 95.1% cross-validation accuracy for high/low sales classification on Dataset D2.
- Built a Streamlit dashboard integrating sales predictions, classification results and association-rule insights to support data-driven inventory and waste-reduction decisions.

Smart Home Environmental Monitoring System* | *IoT
- Contributed to a prototype environmental monitoring system integrating an ESP32, DHT11, MQ-2, MQ-7 and rain sensors to monitor temperature, humidity, smoke, carbon monoxide and rainfall.
- Contributed to embedded C++ functionality for hazard detection, buzzer and email alerts, and servo-controlled laundry retraction based on environmental thresholds.
- Collaborated on a Spring Boot dashboard with REST APIs, HTTPS and cloud database integration for environmental monitoring, historical data visualisation and remote actuator control.

Smart Transportation Tracking System* | *Edge Computing & IoT
- Integrated GPS and MPU-6050 sensors with an ESP32 to monitor vehicle location, speed and motion.
- Implemented local C++ edge processing to filter sensor noise and detect harsh braking, rapid acceleration, sharp turns and overspeeding.
- Integrated Blynk IoT for real-time monitoring and adaptive data transmission to reduce unnecessary cloud communication while prioritising critical events.

Intrusion Detection & Prevention Lab* | *Snort, Suricata, Wazuh, Splunk

- Configured a VMware-based cybersecurity lab with Kali Linux, Ubuntu NIDS, Windows 7 and Metasploitable 2 for network segmentation, attack simulation and security monitoring.
- Implemented Snort and Suricata for network intrusion detection and prevention, alongside Wazuh HIDS for Windows file, registry, user account, login and event-log monitoring.
- Integrated security logs with Wazuh and Splunk to centralise alerts and investigate events involving Nmap reconnaissance and Windows endpoint activity..

Java Chat Application | Java, Socket Programming, GitHub

- Collaborated on a Java chat application supporting real-time text and image exchange between multiple users.
- Implemented client–server communication using socket programming and managed source code through GitHub.

Database-Driven Crowdfunding Platform (in progress) | Oracle DB, SQL, MVC

- Developing a full-stack crowdfunding platform that enables users to create, manage and promote campaigns for financially constrained projects.
- Integrating Oracle Database and SQL with a frontend through MVC architecture, applying database normalisation, query optimisation and user authentication.

Embedded Data Acquisition Simulation | TinkerCAD

- Simulated a sensor-based data acquisition system in TinkerCAD to demonstrate hardware–software interaction in embedded systems..

Network Connectivity Analysis | PingPlotter

- Used PingPlotter to analyse local and global network paths, identify latency and packet-loss patterns, and investigate potential network bottlenecks.

SULAM AI Outreach Program | AI/ML Workshops

- Supported the planning and facilitation of AI/ML workshops for secondary school students by coordinating logistics, learning materials and scheduling.

EDUCATION

Bachelor of Computer Science with Honours, Universiti Putra Malaysia, Selangor | *Oct 2023 – Present*
Current CGPA: 3.9
Core Computer Science & Systems, Software Engineering & Development, AI & Data Analytics, Cybersecurity & Networks, Embedded Systems & IoT, and Computing Ethics & Business

Foundation in Agriculture Science, Universiti Putra Malaysia | *GPA: 4.00*

AWARDS & CERTIFICATIONS

- Gold Medal, PEDi COLLOQUIUM 2026 (G2G Project)
- Gold Medal, PIITRAM 2023 Innovation Competition (PIITRAM)
- Information Literacy Programme (UPM)
- Figma UI/UX Design (Google Developer Group)
- Mobile App with AI-Integrated Features (Google Developer Group)
- Web Security: Threats, Defenses & Best Practices (Google Developer Group)
- deVMatch Hackathon (APUBCC)
- Program AI Summer Camp (IVI,FCoEE,UKM)
- NexG GODAMLah 2.0 Hackathon (Kolaxus)
- Pitching IoT Incubator Top Agropreneur 2.0 Competition(KPT, CEM UPM)
- Program Hatchquest Startup Bootcamp (UPM)
- Google Cybersecurity Certificate (Google)
- Data Analyst Certificate (PutraMOOC)

LEADERSHIP & VOLUNTEERING

- Committee Member in CSC Club: supported student welfare and school-wide events.
- Auditor, School Youth Cadet: verified financial records, budgets and expenditure reports.
- Secretary, Career Club: recorded minutes and coordinated career events and speaker sessions.
- Event Assistant, SUKMA XIX Perak 2018: performed during the opening ceremony.
- Event Volunteer: Twincity Marathon:staffed hydration stations and supported post-event cleanup.

REFERENCES

Prof. Madya Datin Dr. Norwati Mustapha, Universiti Putra Malaysia (Intelligent Computing, Data Mining)
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